EXPLORATORY DATA ANALYSIS REPORT

Mental Health in Tech

OSMI 2014 Survey · Treatment-Seeking Behavior Among Tech Employees

HR & People Analytics

1,259 respondents · 27 survey variables · 23 visualizations

1. Executive Summary

This report analyzes the 2014 OSMI (Open Sourcing Mental Illness) Mental Health in Tech Survey, covering 1,259 responses from tech-industry employees. The analysis examines how demographics, workplace characteristics, and mental-health support policies relate to whether employees actually seek treatment for a mental health condition.

> **Bottom line:** Treatment-seeking is driven far more by **awareness and communication of existing workplace support** (care options, benefits clarity, anonymity assurances) and by **personal/behavioral signals** (family history, degree of work interference) than by demographics like age or remote-work status. Gender shows a meaningful gap, with male respondents seeking treatment less often than female and non-binary respondents.

2. Problem Statement & Business Objective

Problem: Mental health remains under-discussed in tech workplaces despite high-pressure conditions. Employers struggle to know whether their benefits and policies (health coverage, wellness programs, anonymity guarantees, care-option awareness) actually translate into employees seeking treatment, and whether fear of stigma undermines those efforts.

This analysis was guided by four questions:

- What proportion of tech employees have sought treatment for a mental health condition, and how does this vary by demographic and workplace factors?
- Which workplace policies/attitudes (benefits, care options, anonymity, leave policy, fear of consequences) are most strongly associated with treatment-seeking behavior?
- Does company size, remote-work status, or employer type meaningfully affect the availability of mental health support?
- Are there patterns that could help a company identify where its mental health support is falling short?

Business Objective: Help HR / People teams identify which workplace factors most strongly influence whether employees seek mental health treatment, so they can prioritize investments (clearer communication of benefits, anonymity guarantees, manager training) most likely to increase treatment-seeking and reduce stigma.

3. Dataset Overview & Data Quality

The dataset contains **1,259 rows and 27 columns**, mostly categorical self-reported survey answers, with Age as the only true numeric column. There are no duplicate rows. Missingness is uneven:

Column	Missing %	Reason
comments	~87%	Optional free-text field
state	~40%	Only applies to US respondents
work_interfere	~21%	Skipped by respondents without a mental health condition (informative, not random)
self_employed	~1.4%	Likely skipped or left blank

The Age and Gender columns also contained clear data-entry errors and inconsistencies requiring cleaning.

Key Variables

Variable	Description
treatment	Whether the respondent sought mental health treatment (key outcome)
family_history	Family history of mental illness (Yes/No)
work_interfere	How much a condition interferes with work
no_employees	Company size bucket
benefits / care_options / wellness_program / anonymity / leave	Employer-provided support and access to it
mental_health_consequence	Perceived negative consequences of disclosure
Gender / Age / Country	Demographics

Data Cleaning Performed

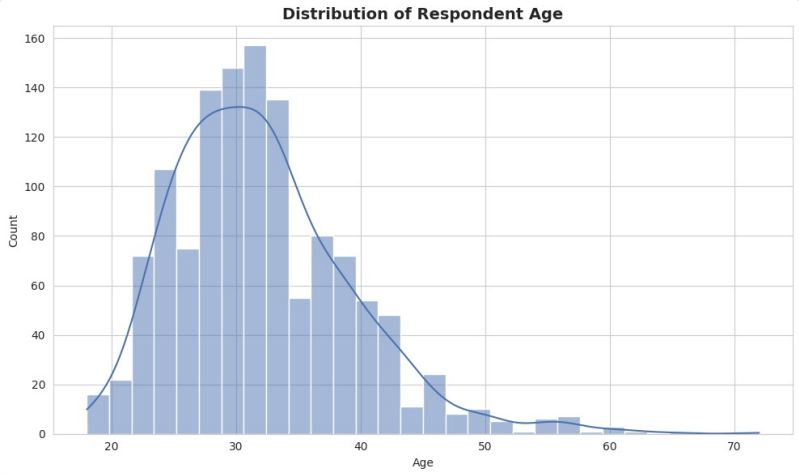
1. Parsed and dropped `Timestamp` (survey period not relevant to this analysis).
2. Replaced invalid `Age` values (e.g. negative numbers, `-1726`, `99999999999`) with the median age (~31).
3. Standardized `Gender` from 40+ free-text variants into three categories: Male, Female, Other/Non-binary.
4. Filled `self_employed` missing values with "No" (majority class) and `work_interfere` missing values with "Not applicable" (per survey skip-logic).
5. Dropped `comments` and `state` due to high, largely irrelevant missingness.

After cleaning, the sample skews heavily male (~79%), consistent with tech-industry demographics, and age is concentrated in the 25–40 range.

4. Visual Analysis

The following 23 charts follow a Univariate → Bivariate → Multivariate structure: first examining individual variables, then their relationship with treatment-seeking, and finally combined multivariate patterns, closing with a correlation heatmap and pair plot.

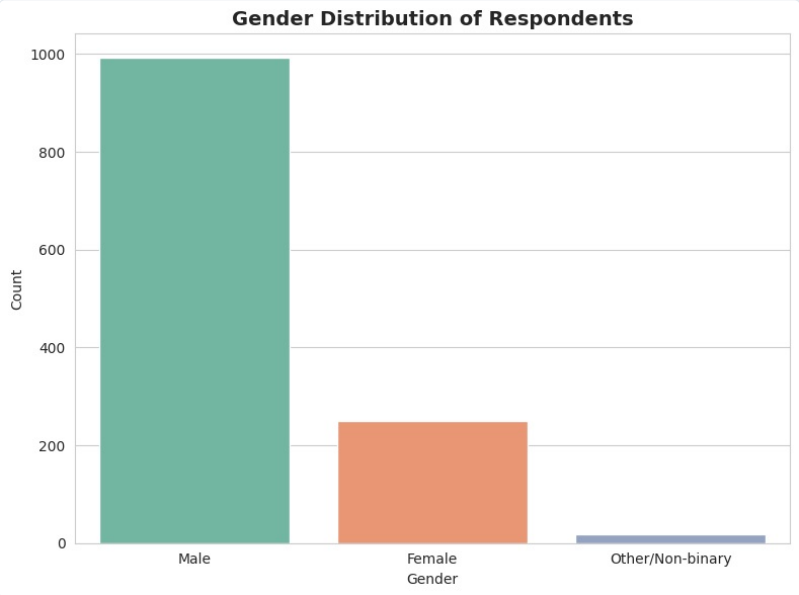
Chart 1. Age Distribution



Insight: The distribution is right-skewed and concentrated between 25-40 years, peaking around 28-32, consistent with a young tech workforce. After cleaning, there are no more unrealistic outliers.

Business Implication: Understanding the age skew helps target mental-health programs (e.g., stress/burnout-specific programs for the dominant 25-35 age group) rather than a one-size-fits-all approach. No negative-growth insight here — it's purely descriptive.

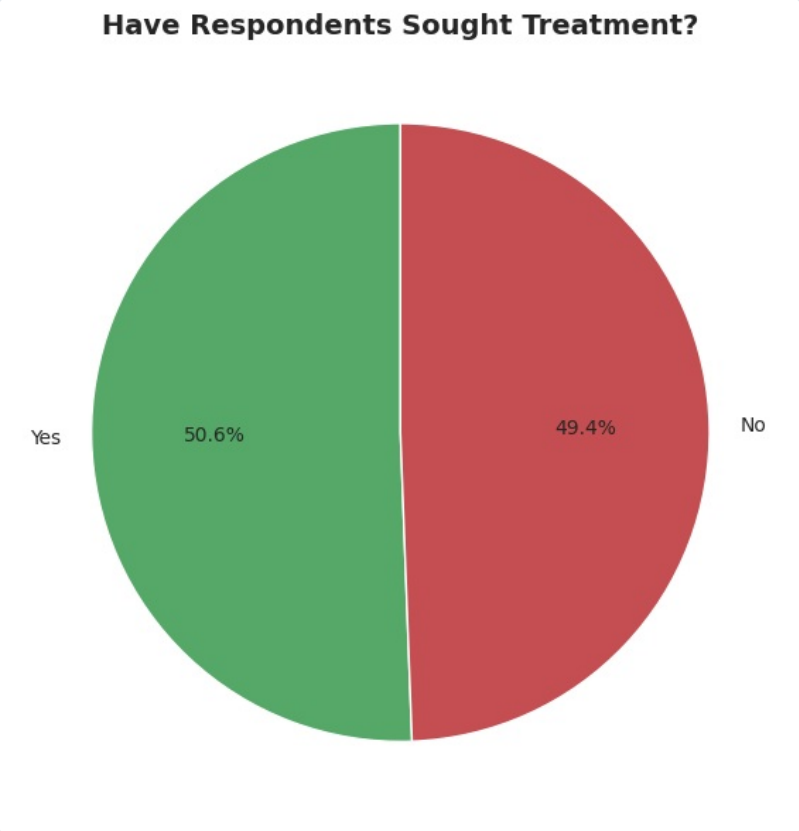
Chart 2. Gender Distribution



Insight: The sample is heavily male (~79%), followed by Female (~19%) and a small Other/Non-binary group (~2%), reflecting known gender imbalance in the tech industry.

Business Implication: This imbalance means any gender-based conclusions (especially for the Other/Non-binary group) should be treated with caution due to small sample size, even though the pattern itself (male-dominated tech workforce) is a well-established finding, not a new risk.

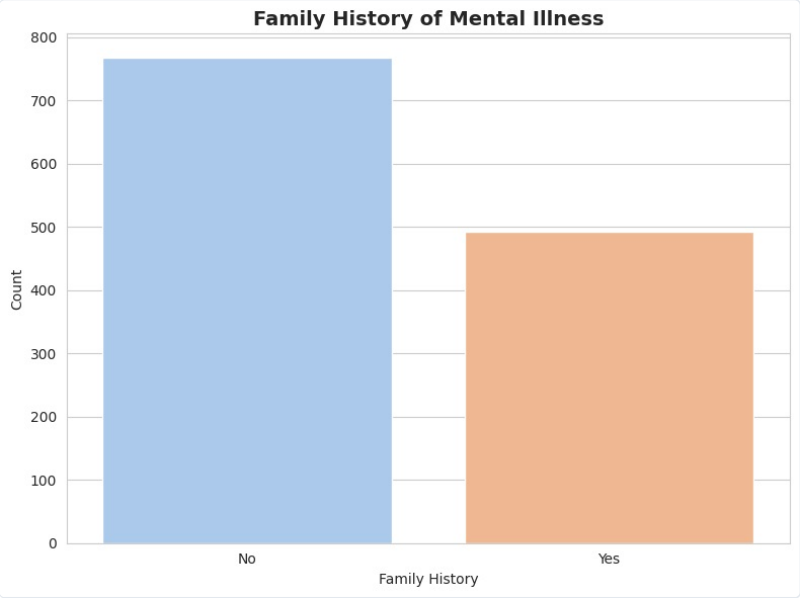
Chart 3. Treatment Sought Distribution



Insight: Roughly half (50.6%) of respondents have sought treatment for a mental health condition, and half have not — an almost even split.

Business Implication: This near 50/50 split is a strong signal that treatment-seeking is not universal even when a condition may be present, reinforcing the business case for improving workplace support to shift more employees toward seeking help.

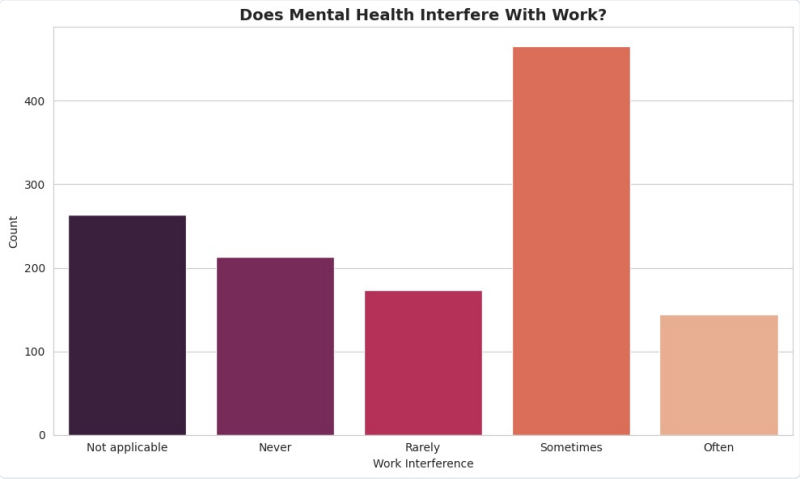
Chart 4. Family History Distribution



Insight: About 39% of respondents report a family history of mental illness, while 61% do not — a substantial minority that is useful context for the later bivariate analysis against treatment.

Business Implication: No direct negative-growth implication; this is a demographic baseline used to interpret later charts.

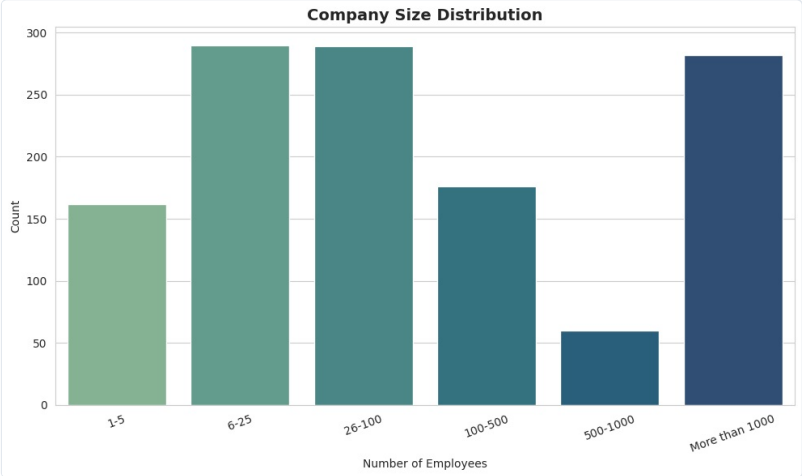
Chart 5. Work Interference Distribution



Insight: Excluding 'Not applicable' (no condition), 'Sometimes' is the most common response, followed by 'Rarely', suggesting mental health conditions intermittently affect work for most affected employees rather than constantly.

Business Implication: Employers could use this to justify flexible/adaptive work policies (e.g., flexible hours on 'bad days') rather than blanket accommodations, since interference is often intermittent rather than constant.

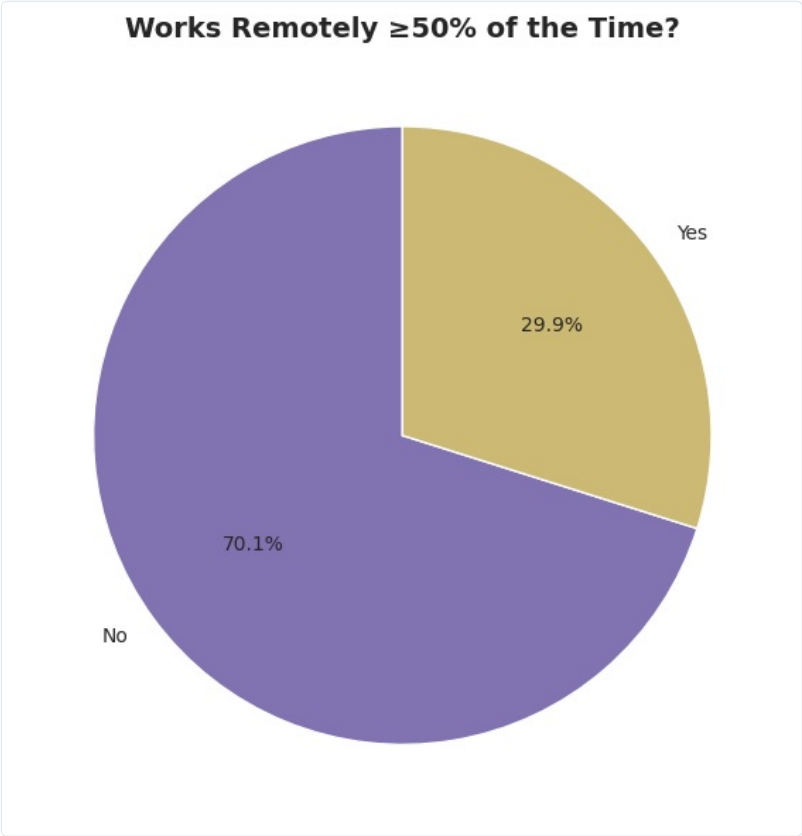
Chart 6. Company Size Distribution



Insight: Respondents are fairly evenly spread across company sizes, with a slight concentration in 'More than 1000' employees and '26-100', indicating the survey captured both large enterprises and small startups.

Business Implication: A diverse company-size sample makes it easier to generalize benefit-availability findings later across small and large employers rather than being biased toward one segment.

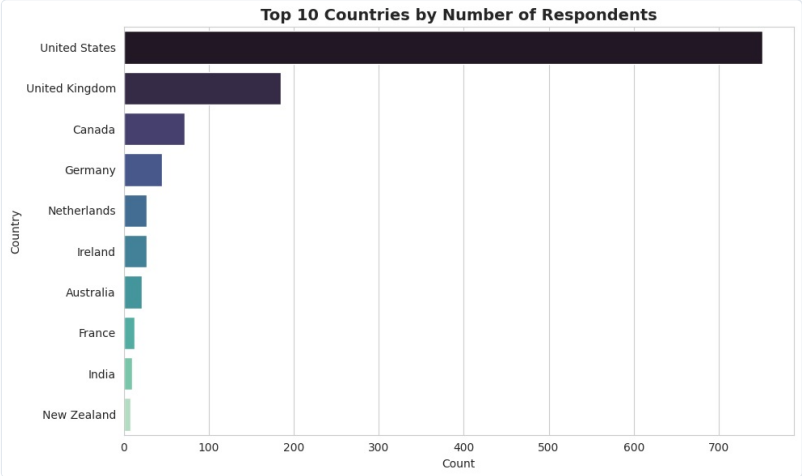
Chart 7. Remote Work Distribution



Insight: Only about 30% of respondents work remotely at least half the time — in this 2014 dataset, in-office work was still the norm.

Business Implication: This is a useful baseline: any relationship found later between remote work and treatment-seeking should be interpreted knowing remote workers are a minority in this sample.

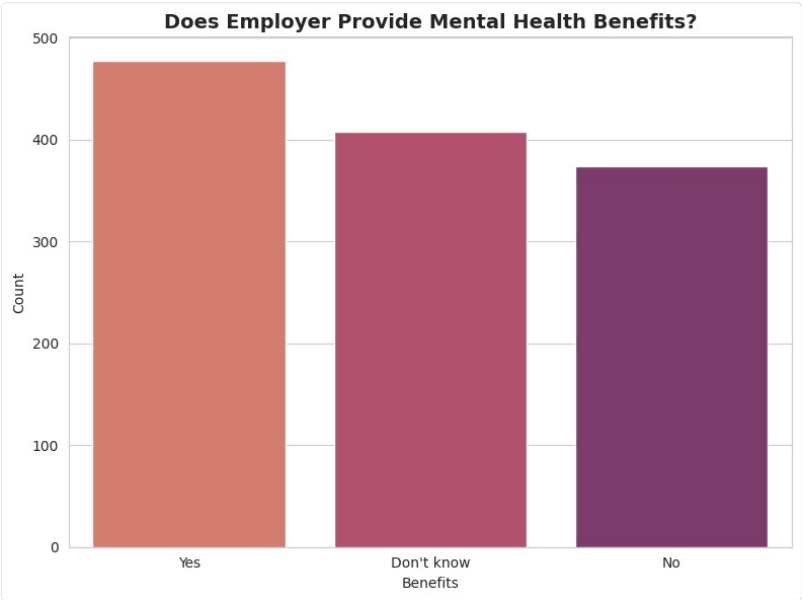
Chart 8. Top 10 Countries by Respondent Count



Insight: The United States dominates the sample (~60%), followed distantly by the United Kingdom and Canada, meaning the findings mostly reflect US workplace norms and healthcare context.

Business Implication: This is an important caveat: recommendations drawn from this dataset may not generalize well to countries with different healthcare systems (e.g., universal healthcare countries), and should be validated locally before being applied globally.

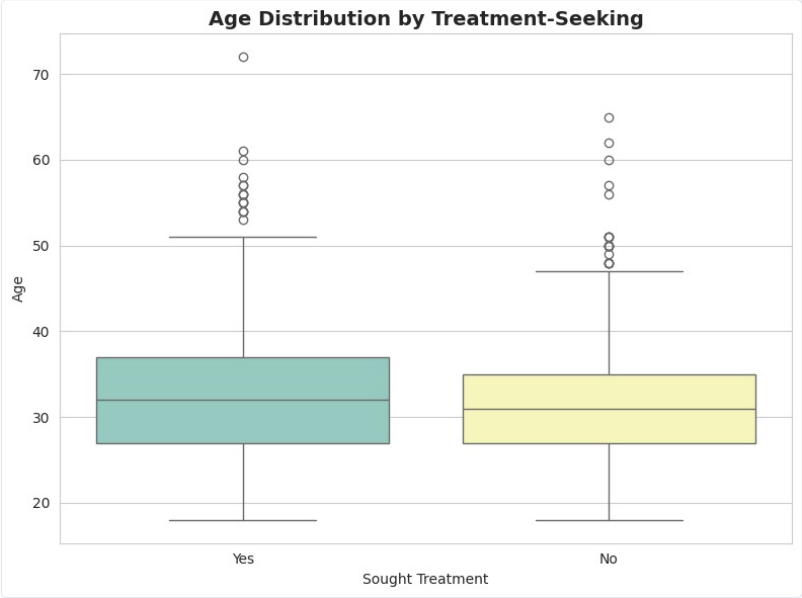
Chart 9. Mental Health Benefits Availability



Insight: Only about 39% of respondents confirm their employer provides mental health benefits, while a notable ~29% simply 'don't know' — almost as many as those who said 'No'.

Business Implication: The large 'Don't know' segment is itself a finding: it suggests a communication gap, not necessarily a lack of benefits. This is a low-cost, high-impact opportunity — improving internal communication about existing benefits could move some of that 29% into confirmed awareness without a policy change.

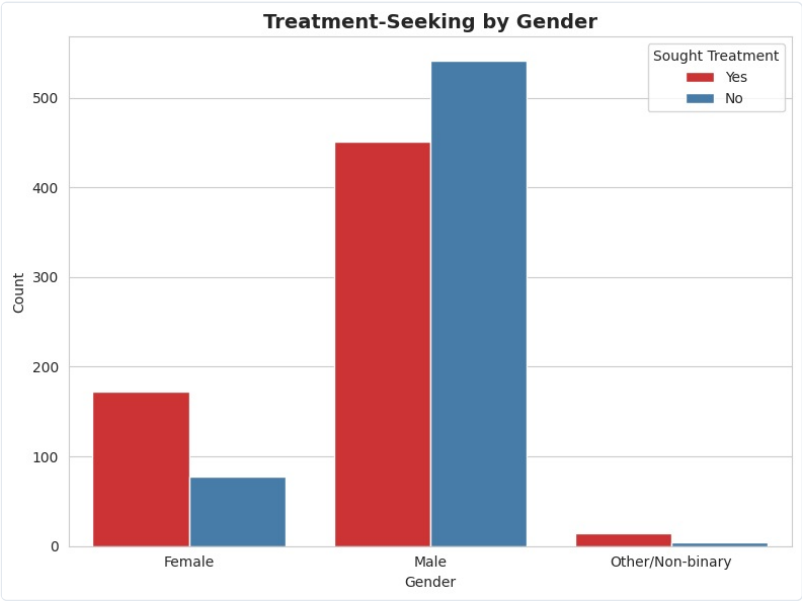
Chart 10. Age vs Treatment



Insight: Median age is nearly identical between the two groups, with only a slightly higher median for those who sought treatment — age alone is not a strong differentiator of treatment-seeking behavior.

Business Implication: Since age is not a strong driver, mental-health outreach programs should not be age-targeted; broader, company-wide policies are more likely to be effective than age-specific initiatives.

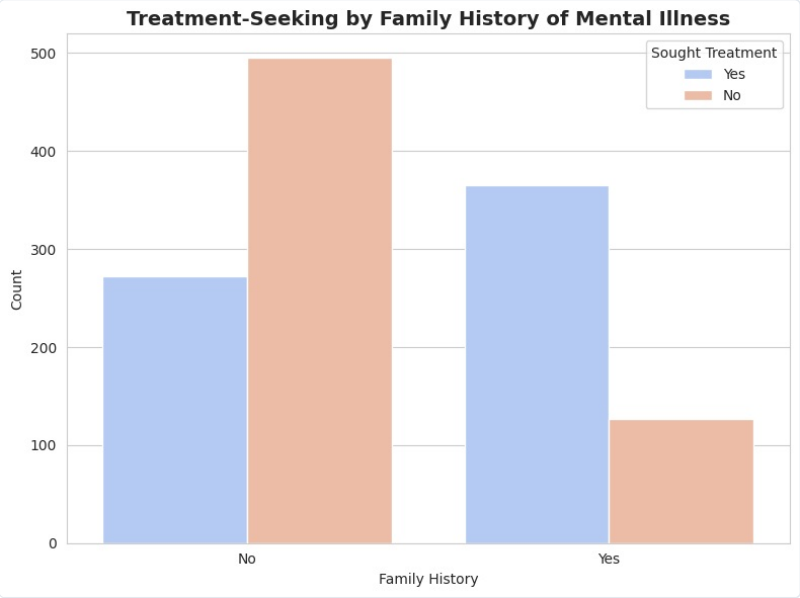
Chart 11. Gender vs Treatment



Insight: Female and Other/Non-binary respondents seek treatment at a notably higher rate than Male respondents, where the split is closer to even.

Business Implication: This suggests outreach and stigma-reduction efforts may need to be especially targeted at male employees, who appear less likely to seek treatment relative to their peers — a potential negative-growth risk if untreated conditions among men go unaddressed and affect long-term productivity/retention.

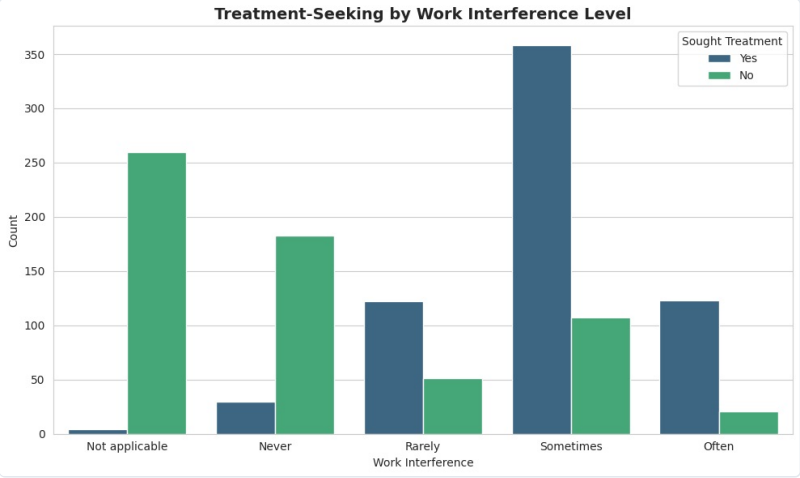
Chart 12. Family History vs Treatment



Insight: Respondents with a family history of mental illness are far more likely to have sought treatment (~75%) compared to those without (~35%) — one of the strongest relationships in the dataset.

Business Implication: This is a strong positive insight: family history is a practical, low-friction signal that could be used in **opt-in** wellness screening/education programs, e.g., proactively offering information to employees who self-identify a family history, without being invasive.

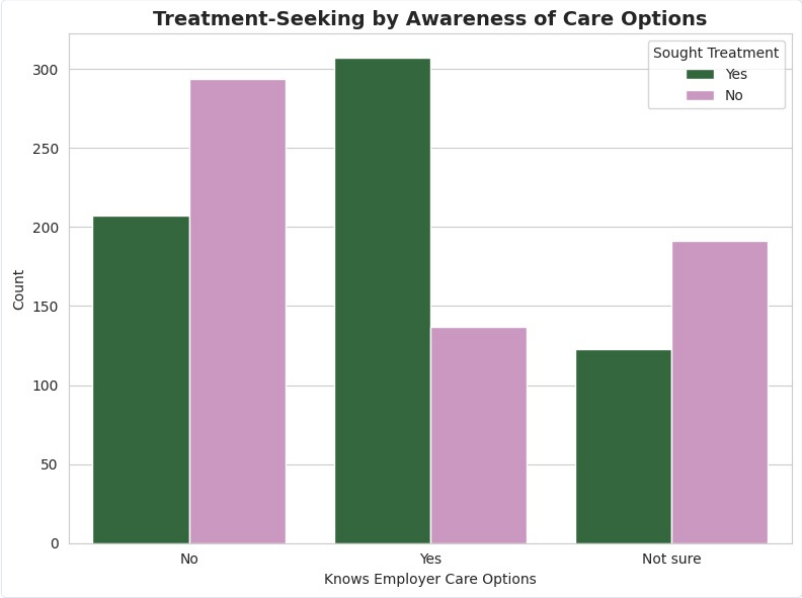
Chart 13. Work Interference vs Treatment



Insight: Treatment-seeking rises sharply with interference severity — nearly all respondents reporting 'Often' interference have sought treatment, while very few reporting 'Never' have. This confirms work_interfere as the strongest behavioral predictor of treatment in the dataset.

Business Implication: This is highly actionable: managers could be trained to recognize early, mild interference signals ('Rarely'/'Sometimes') and proactively point employees to resources, rather than waiting until interference becomes severe ('Often'), potentially reducing burnout-driven attrition.

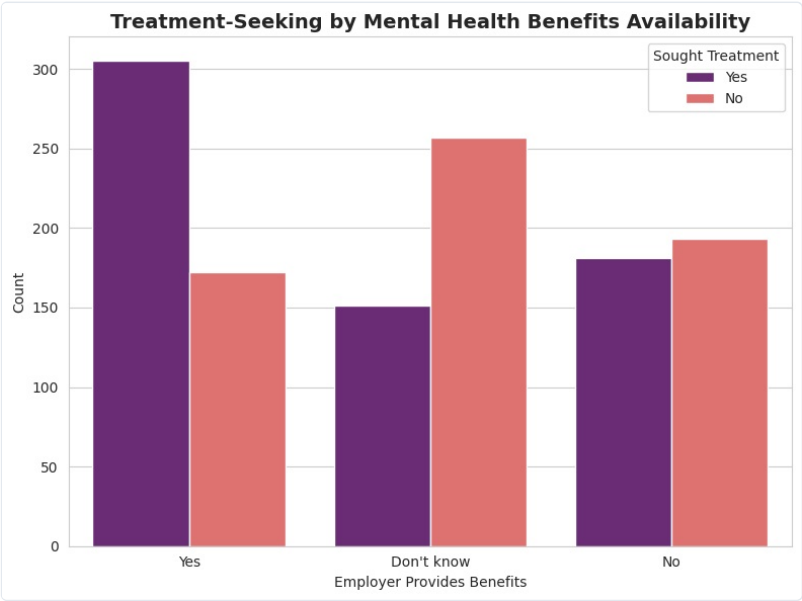
Chart 14. Care Options Awareness vs Treatment



Insight: Employees who know their employer's mental health care options seek treatment at a much higher rate (~65%) than those who say 'No' (~40%) or 'Not sure' (~35%).

Business Implication: This directly supports investing in clearer communication of existing care options — a relatively low-cost intervention (e.g., onboarding materials, regular reminders) that appears strongly associated with more employees getting help.

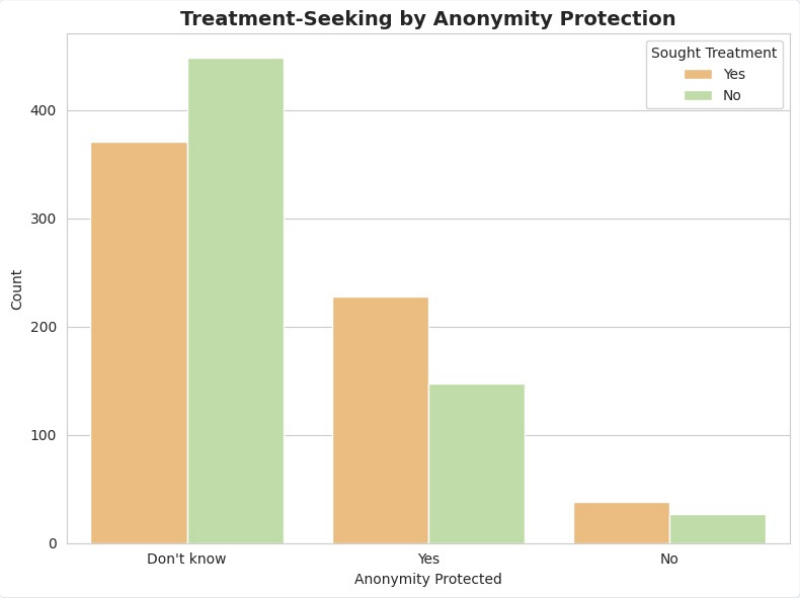
Chart 15. Benefits vs Treatment



Insight: Respondents whose employer provides benefits seek treatment more often (~63%) than those who say 'No' (~45%) or 'Don't know' (~44%) — confirming that formal benefits do correlate with treatment, though the gap is smaller than for care-option awareness.

Business Implication: Positive impact: expanding benefits access could help, but the smaller gap (vs. care_options) suggests that simply offering benefits without ensuring employees know about and understand them yields diminishing returns — awareness may matter more than the benefit itself.

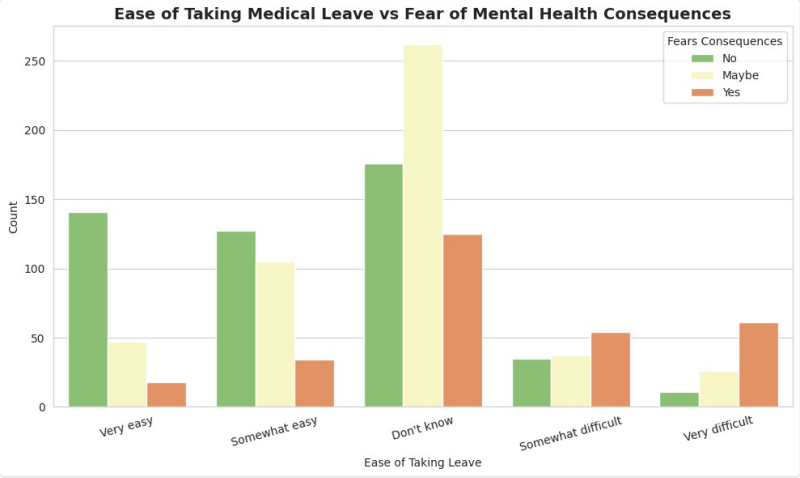
Chart 16. Anonymity Protection vs Treatment



Insight: The 'Yes' group has a noticeably higher treatment rate than the large 'Don't know' group, which is the majority response — most employees are simply unsure whether their anonymity is protected.

Business Implication: The dominance of 'Don't know' responses (rather than 'No') again points to a communication gap rather than a policy gap — clearly and repeatedly communicating anonymity guarantees could be a quick win to boost treatment-seeking confidence.

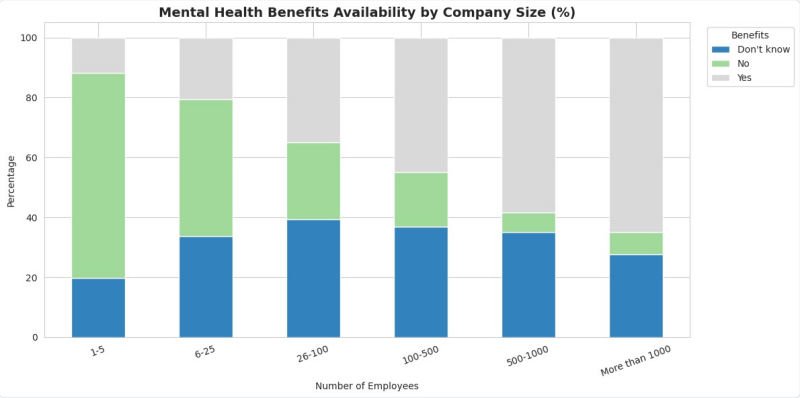
Chart 17. Ease of Leave vs Fear of Mental Health Consequences



Insight: Respondents who find leave 'Very difficult' or 'Somewhat difficult' report fearing consequences far more often than those who find it 'Very easy' — and a large 'Don't know' group again dominates the leave-ease question.

Business Implication: This links two policy levers together: making leave policies clear and genuinely easy to use appears associated with less fear of stigma. Ambiguous leave policies (the large 'Don't know' segment) may be silently discouraging people from taking needed time off — a potential negative-growth risk if unaddressed (burnout, presenteeism).

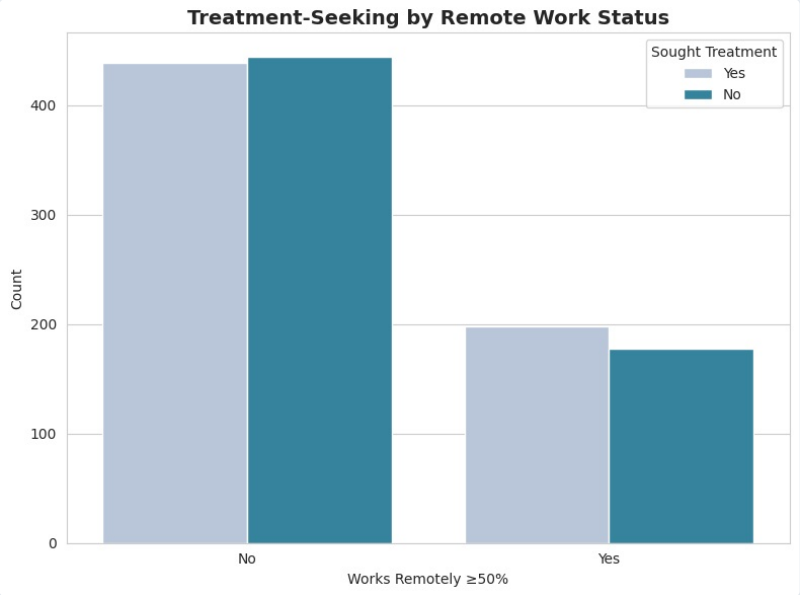
Chart 18. Company Size vs Benefits Availability



Insight: Larger companies (500+ employees) are far more likely to formally provide mental health benefits, while small companies (1-5, 6-25 employees) show much higher 'No' and 'Don't know' shares.

Business Implication: This is an important equity insight: employees at small/early-stage companies are structurally disadvantaged in mental health support. For a business audience, this means startups may need to consider low-cost alternatives (EAPs, stipends) since traditional benefit packages scale with company size.

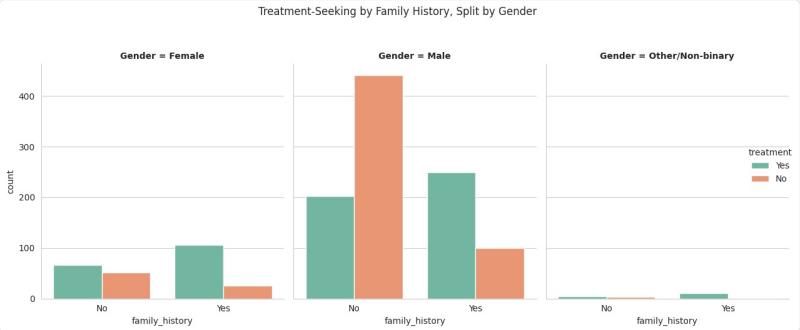
Chart 19. Remote Work vs Treatment



Insight: Treatment-seeking rates are almost identical between remote and non-remote workers, indicating remote work status alone doesn't meaningfully influence treatment-seeking in this sample.

Business Implication: No strong actionable signal here — this suggests companies shouldn't assume that remote-work policy changes alone will affect mental-health treatment rates; other factors (stigma, awareness) matter more.

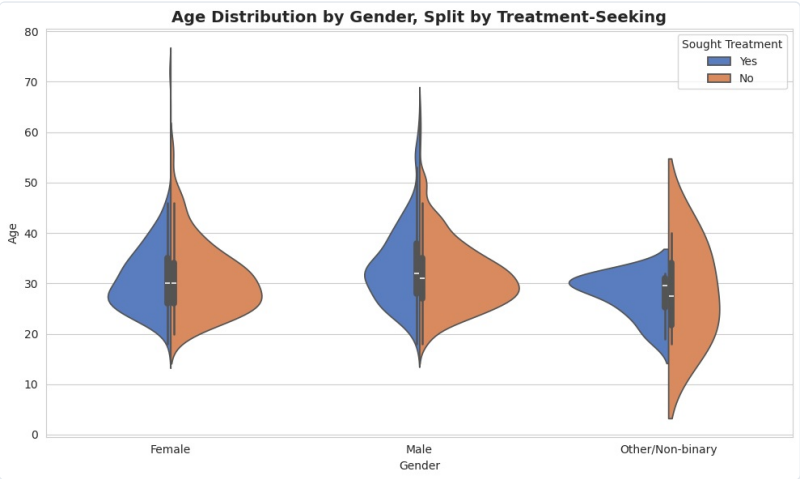
Chart 20. Gender, Family History & Treatment (Multivariate)



Insight: The strong 'family history → treatment' relationship holds within every gender group, but the gap between 'has family history' and 'no family history' is widest for Male respondents, suggesting family history has an especially strong influence on whether men seek treatment.

Business Implication: This refines the earlier gender insight: outreach to male employees could be specifically framed around family-history awareness (e.g., 'if mental illness runs in your family, here's what support is available'), which may resonate more than generic messaging.

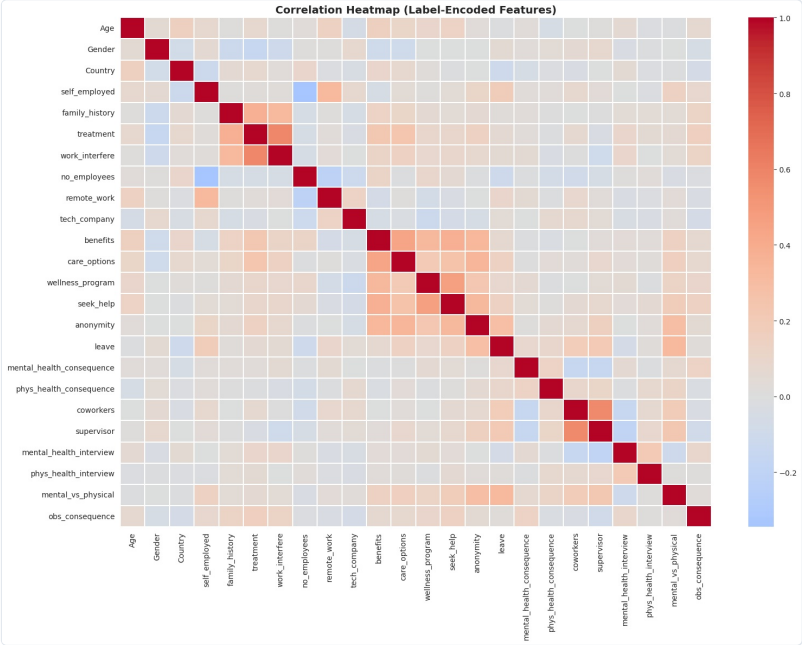
Chart 21. Age Distribution by Gender, Split by Treatment (Multivariate)



Insight: Age distributions are broadly similar across gender and treatment groups, reinforcing that age is not a major differentiator, while gender-based differences (seen in Chart 11) hold across the age range rather than being concentrated in a specific age band.

Business Implication: Confirms that interventions should be designed around gender/awareness factors rather than age segments, consistent with the age-related insight from Chart 10.

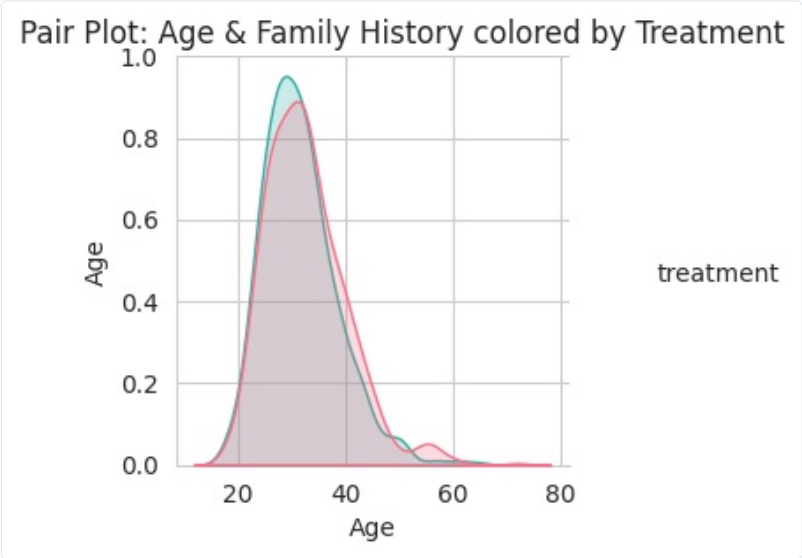
Chart 22. Correlation Heatmap



Insight: `treatment` shows its strongest correlations with `family_history`, `work_interfere`, and `care_options`, consistent with the bivariate findings above. Most other workplace-policy variables show weak-to-moderate correlations with each other (e.g., `benefits`, `wellness_program`, `seek_help` cluster together), suggesting companies that offer one form of support tend to offer several.

Business Implication: This validates that `family_history`, `work_interfere`, and `care_options` are the top three factors to prioritize when designing interventions — a data-driven way to focus limited HR resources rather than spreading efforts thin across all 20+ policy variables.

Chart 23. Pair Plot



Insight: There's no strong visual separation by age alone (confirming Chart 10/21), and the KDE diagonal shows treatment-seeking respondents are spread across the same age range as non-seekers — age simply doesn't segment the outcome well in this dataset.

Business Implication: Reinforces (rather than reveals new) findings: with only one true numeric variable, deeper multivariate structure lives in the categorical variables (family_history, work_interfere, care_options) rather than in Age.

5. Recommendations

1. **Prioritize communication over new policy creation.** A large share of respondents answered "Don't know" — not "No" — to benefits, anonymity, and care-option questions. This is the single biggest, lowest-cost opportunity: run regular, clear communication campaigns about existing mental health resources.

2. **Train managers to recognize early work-interference signals.** Treatment-seeking rises sharply with interference severity, so coaching managers to notice early signs ("Rarely"/"Sometimes") can prompt earlier intervention.
3. **Make care-option and benefit information genuinely easy to find** — awareness of care options showed a stronger relationship with treatment-seeking than the mere existence of benefits.
4. **Address the gender gap directly** with messaging that acknowledges men are statistically less likely to seek treatment, favoring anonymous/no-stigma channels first.
5. **Support small companies differently** — since benefit availability scales with company size, startups should consider low-cost alternatives (EAPs, mental-health stipends, telehealth partnerships).
6. **Simplify and clarify medical leave policy for mental health**, since unclear/difficult leave policies correlate with higher fear of disclosure consequences.

> **Caution:** The dataset is heavily skewed toward the US (~60%) and male respondents (~79%). Recommendations should be validated against a specific company's own demographic and country mix before being applied — over-generalizing US-centric findings to different healthcare/labor contexts could be counterproductive.

6. Conclusion

This EDA shows that whether a tech employee seeks treatment is driven far more by **awareness and communication of workplace support** and by **personal/behavioral signals** (family history, work interference) than by demographic factors like age or remote-work status. Gender shows a meaningful gap, with male respondents seeking treatment less often.

Across the univariate, bivariate, and multivariate analysis — backed by a correlation heatmap and pair plot — three variables consistently stood out as the strongest correlates of treatment-seeking: **family_history**, **work_interfere**, and **care_options**. This gives HR teams a clear, data-driven starting point for where to focus mental health support investments.

Source: OSMI Mental Health in Tech Survey (2014), 1,259 respondents. Report generated from exploratory data analysis notebook.